IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received: June 23, 2026; Revised: –; Accepted: –

*Digital Object Identifier 10.1109/ACCESS.2026.XXXXXXX*

# FoodSense: A Multimodal IoT–AI Framework for Early Food Spoilage Detection via On-Edge Drift-Compensated Gas Sensing, Virtual Visible Spectroscopy, and Deep Visual Analysis

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This work was carried out as an Interdisciplinary Project at RV College of Engineering, Bengaluru, during the academic year 2025–2026. No external funding was received.

**ABSTRACT** Food spoilage is responsible for approximately 19% of global food production loss, equivalent to 1.3 billion tonnes annually, with cascading consequences for food security, greenhouse gas emissions, and economic stability. Existing automated detection approaches suffer from fundamental limitations: commercial electronic noses employ large metal-oxide sensor arrays costing several thousand USD; deep learning systems operating on visual data alone neglect biochemical precursors that precede visible decay; and available systems universally report a static binary classification rather than a continuous estimate of remaining shelf life. This paper presents **FoodSense**, a comprehensive multimodal sensing and inference framework that addresses all three gaps simultaneously. The system integrates three orthogonal sensing modalities: (i) an MQ135 metal-oxide gas sensor with closed-form temperature-humidity drift compensation executed in real-time on an Arduino ATmega328P microcontroller; (ii) a DIY RGB-LED/LDR visible spectrophotometer with a machine-learning virtual calibration layer that maps raw photodetector signals to continuous wavelengths and Beer-Lambert absorbances; and (iii) a deep visual analysis branch employing a fine-tuned EfficientNet-B0 convolutional neural network and a Vision Transformer (ViT-B/16) for produce-image spoilage assessment. On the UCI Gas Sensor Array Drift Dataset (13,910 samples), a batch-normalised Multi-Layer Perceptron achieves **96.84%** accuracy and an F1-score of **87.28%**. An attention-augmented Long Short-Term Memory network provides continuous shelf-life regression with a mean absolute error of ±**4.0 hours** on a proof-of-concept synthetic time-series. A multi-task calibration MLP maps RGB-LDR signals to visible wavelengths with $R^2 = 0.9625$, and a pigment-degradation classifier resolves Fresh, Ripe, and Spoiled stages at **95.60%** accuracy by exploiting chlorophyll and enzymatic-browning absorption signatures.

**INDEX TERMS** food spoilage detection, IoT sensor fusion, metal-oxide gas sensor, drift compensation, MLP classifier, attention-augmented LSTM, visible spectroscopy, virtual spectrophotometer, Vision Transformer, shelf-life prediction, multimodal deep learning, edge inference, beer-lambert law, chlorophyll degradation

## I. INTRODUCTION

FOOD waste constitutes one of the defining challenges of the 21st century. The United Nations Food and Agriculture Organisation (FAO) estimates that approximately 1.3 billion tonnes of food—roughly 19% of all food produced globally—is lost or wasted annually before reaching consumers [1]. A disproportionate share of this loss occurs during post-harvest storage, cold-chain transit, and retail holding, all stages at which early detection of spoilage onset could dramatically extend edible shelf life and reduce discard rates.

Beyond food security, rotting organic matter contributes an estimated 8–10% of global greenhouse gas emissions as it decomposes in landfills, reinforcing the environmental urgency of the problem.

Spoilage is a fundamentally biochemical process. Microbial colonisation triggers the release of volatile organic compounds (VOCs) including ammonia ($NH_3$), hydrogen sulfide ($H_2S$), trimethylamine (TMA), and ethylene ($C_2H_4$), collectively responsible for the characteristic odour of decaying food. Simultaneously, enzymatic pathways degrade chlorophyll (*a* and *b*) in plant tissue through the action of chlorophyllase and pheophytinase, reducing green optical reflectance. Polyphenol oxidase (PPO) and peroxidase (POD) catalyse the oxidation of phenolic substrates to dark-coloured melanin quinones, producing visible browning at wavelengths centred near 540 nm. These biochemical signatures—VOC emission, chlorophyll loss, and browning pigment accumulation—evolve on different timescales and are therefore complementary: VOCs provide the earliest sub-threshold warning, chlorophyll loss tracks intermediate degradation, and browning correlates with terminal spoilage.

The commercial solution—an electronic nose (e-nose) built from a calibrated array of 16 or more metal-oxide semiconductor (MOS) sensors with specialised signal conditioning and laboratory-grade drift correction—costs upward of several thousand USD, demands expert operation, and occupies a benchtop instrument footprint [2]. Consumer-grade alternatives are confined to single-modality solutions: smartphone-based colour analysis [3] ignores olfactory signals entirely, while single MOS sensors lack the multi-species discrimination of full e-nose arrays and suffer from well-documented baseline drift driven by ambient temperature and humidity fluctuations [4]. Critically, all of these approaches output a point-in-time classification (“fresh” or “spoiled”) rather than a continuous, dynamically updated estimate of remaining shelf life—the quantity that is most actionable for supply-chain management and consumer decision-making.

A survey of the recent literature reveals three persistent and intersecting gaps:

1) **Single-modality limitation.** The majority of published food-quality systems exploit either gas-phase VOC sensing *or* visual analysis, but not both. Di Rosa et al. [5] showed theoretically that fusing an electronic nose, electronic tongue, and computer vision dramatically improves discrimination accuracy across food-type boundaries, yet low-cost implementations that realise all three modalities remain absent from the literature.
2) **Absence of continuous temporal forecasting.** Static classifiers report an instantaneous binary or ordinal label. Spoilage, however, is a kinetic process following approximately first-order microbial growth dynamics. The output that matters operationally is not “is this food currently spoiled?” but “how many hours of safe shelf life remain?” This requires modelling the temporal trajectory of VOC signals, not just their current level.
3) **Offline-only drift compensation.** Metal-oxide sensors undergo baseline drift of up to 30% in resistance within the first year of deployment [2]. Most published systems compensate for this drift through batch post-processing in laboratory settings, making them unsuitable for unattended embedded deployment where no human recalibration is possible.

**Contributions.** This paper presents FoodSense, a multimodal IoT–AI framework that addresses all three gaps through the following specific technical contributions:

1) **Real-time on-edge drift compensation** (§III): A closed-form temperature-humidity correction formula is derived and implemented in firmware on the Arduino ATmega328P (16 MHz, 32 KB Flash), requiring fewer than 8 bytes of RAM and executing in under 1 ms—well within the sensor sampling budget.
2) **Attention-augmented LSTM for continuous shelf-life regression** (§V): A 2-layer LSTM with a learned temporal attention mechanism processes a 30-reading (1-hour) rolling context window and outputs a real-valued estimate of remaining shelf life in hours, superseding the conventional binary label.
3) **Sub-$5 virtual visible spectrophotometry** (§VI): A multi-task MLP calibration model maps raw RGB-LED/LDR photodetector signals to continuous visible wavelengths (nm) and Beer-Lambert absorbances, realising a functional visible spectrometer from commodity components at a fraction of the cost of commercial instruments.
4) **Three-branch multimodal inference architecture** (§IV): Gas/climate, spectroscopy, and visual sensing branches operate in parallel, with their outputs fused into a unified freshness score that exploits the complementary temporal and spectral characteristics of each modality.

The remainder of this paper is organised as follows. Section II surveys the relevant literature. Section III describes the hardware sensing platform. Section IV presents the multimodal system architecture. Section V details the AI model designs. Section VI describes the spectroscopy sub-system. Section VII presents and analyses experimental results. Section VIII discusses limitations and future directions. Section IX concludes.

## II. RELATED WORK

### A. GAS SENSOR-BASED SPOILAGE DETECTION AND DRIFT COMPENSATION

Metal-oxide semiconductor gas sensors detect VOCs by measuring the change in resistance of a heated metal-oxide surface (typically $SnO_2$ or $WO_3$) in the presence of reducing or oxidising gases. Their response follows a power-law relationship between sensor resistance and gas concentration, and they are sensitive to a broad spectrum of VOCs simultaneously—a property that makes them both versatile and difficult to calibrate absolutely.

Vergara et al. [2] established the foundational benchmark in this domain by collecting a 36-month gas sensor drift dataset from a 16-sensor MOS e-nose array exposed to ammonia, acetone, and ethylene at known concentrations. The dataset reveals that sensor resistance can drift by over 30% within the first year, driven by ageing of the metal-oxide surface and cumulative exposure to ambient contaminants. Huerta et al. [4] subsequently quantified the separate contributions of ambient temperature and relative humidity to open-sampling e-nose drift, demonstrating that a simple linear correction model derived from environmental measurements substantially reduces drift artefacts in continuous monitoring scenarios. Their work motivates the on-edge temperature-humidity compensation strategy adopted in FoodSense.

Yi et al. [6] proposed a batch drift compensation algorithm based on local discriminant subspace learning, which transfers a classifier trained on early-epoch sensor data to later epochs by learning a shared embedding in which drift is orthogonal to the gas-class discriminant. Their approach achieves state-of-the-art accuracy on the Vergara benchmark but requires labelled target-domain samples—a constraint that precludes deployment in settings where no reference gas samples are available. Chang et al. [7] extended this analysis to temporal and spatial drift artefacts in the same public dataset, identifying specific sensor types that exhibit non-stationary variance patterns not captured by first-order correction models.

For food-specific applications, Guo et al. [8] combined an e-nose with principal component analysis (PCA) and linear discriminant analysis (LDA) to classify *Penicillium expansum* mould and physical defects in apples, achieving overall classification accuracies above 90%. Ricci et al. [9] demonstrated a machine-learning approach applied to microwave impedance sensing for food-quality monitoring, showing that non-optical electromagnetic sensing can extract food quality signals that complement gas and visual modalities.

FoodSense diverges from this body of work by implementing drift compensation as a closed-form formula executed on the sensor node microcontroller itself, requiring no labelled target-domain samples, no post-processing server, and no human intervention once deployed.

### B. VISIBLE SPECTROSCOPY FOR NON-DESTRUCTIVE FOOD QUALITY ASSESSMENT

Non-destructive spectroscopic analysis has been applied to food quality assurance for several decades, predominantly using near-infrared (NIR) and Vis-NIR spectrometers. Fodor et al. [10] reviewed two decades of NIR applications in food quality, identifying key absorption peaks: chlorophyll *a* and *b* absorb strongly at approximately 430 nm and 660–680 nm respectively; carotenoids absorb in the 400–500 nm range; and melanin (enzymatic browning product) absorbs broadly in the 400–600 nm range with a local maximum near 540 nm. These spectral signatures are stable across food types, conferring transferability to new produce categories. Cortés et al. [11] reviewed Vis-NIR monitoring strategies for agricultural products, identifying wavelength ranges that are predictive of moisture content, sugar content, and degradation across fruits, vegetables, and grains.

Consumer-grade spectroscopy implementations have been demonstrated with low-cost components. Albert et al. [12] showed that an Arduino-controlled RGB LED paired with a silicon photodiode can perform quantitative Beer-Lambert absorbance measurements for coloured solutions, achieving accuracy within 5% of a laboratory-grade spectrophotometer for measurements in the visible range. Al-Sabbagh and Abdulrazzaq [13] extended this approach to multi-wavelength visible light measurements for dye concentration analysis. Ren et al. [14] demonstrated a 3D-printed modular spectrometer that enables spectral experiments across the visible range, validating the concept of educational low-cost spectrometry.

FoodSense advances this body of work by introducing a machine-learning calibration layer that transforms the discrete three-channel (RGB) emission profile of a commodity LED into a continuous virtual spectrum, recovering both wavelength and absorbance information that would otherwise require a diffraction grating or filter wheel.

### C. DEEP LEARNING FOR VISUAL FRESHNESS AND RIPENESS ASSESSMENT

Convolutional neural networks have become the dominant approach for visual food quality classification. Amin et al. [3] applied transfer learning with VGG16, ResNet-50, and InceptionV3 architectures to a fruit freshness dataset, demonstrating that fine-tuned ImageNet-pretrained models outperform custom-trained shallow networks due to the richness of low-level visual features learned from large-scale datasets. Their best model achieved high classification accuracy distinguishing fresh and rotten instances of apples, bananas, and oranges.

Gao et al. [15] investigated Vision Transformers for food image classification, demonstrating that self-attention mechanisms capture global texture and colour features that are missed by CNNs with limited receptive fields. ViT architectures trained on augmented food datasets showed superior generalisation to novel lighting conditions and camera viewpoints—a practically important characteristic for consumer deployment. Nguyen et al. [16] designed an optical fruit-ripeness detector combining custom feature extraction with lightweight classification for deployment on embedded hardware in automated fruit-sorting conveyors.

The key limitation of visual-only approaches is their inability to detect sub-threshold biochemical spoilage: VOC emission and chlorophyll degradation begin hours to days before visible colour or texture changes manifest on the produce surface. FoodSense integrates the visual branch as one of three complementary sensing modalities, exploiting visual information for terminal spoilage confirmation while relying on gas and spectroscopic sensing for early detection.

### D. TEMPORAL MODELLING FOR SHELF-LIFE FORECASTING

Long Short-Term Memory networks and attention mechanisms have demonstrated strong performance on food-related time-series prediction tasks. Wu et al. [17] proposed the GCNN-LSTM-AT network for tangerine quality prediction using Vis-NIR spectroscopy, combining graph convolutional networks with LSTM layers and temporal attention to estimate multiple quality attributes simultaneously from time-series spectral measurements. Their work established that temporal context significantly outperforms point-in-time spectral classification for quality regression tasks.

Doganis et al. [18] demonstrated that recurrent neural networks could forecast short shelf-life food product sales from historical time-series data, showing that neural temporal models can capture the kinetic dynamics of food quality evolution. Saha et al. [19] constructed an IoT-based fruit quality inspection and lifespan detection system that used edge-deployed machine learning to combine multiple sensor streams for freshness estimation, validating the concept of embedded AI inference for food quality monitoring.

FoodSense extends temporal modelling to continuous shelf-life regression using a context-window LSTM with learned temporal attention, allowing the model to differentially weight the rapid VOC ramp characteristic of terminal spoilage while suppressing the influence of transient gas-concentration spikes.

## III. HARDWARE SENSING PLATFORM

### A. GAS AND CLIMATE SENSOR NODE

The gas-sensing hardware node is housed in a compact enclosure containing an Arduino Uno microcontroller (ATmega328P, 16 MHz, 32 KB Flash, 2 KB SRAM), an MQ135 metal-oxide gas sensor, and a DHT11 digital temperature-humidity sensor. The node is powered by a 9 V rechargeable battery pack regulated to 5 V by an onboard linear regulator.

**MQ135 Sensor Physics.** The MQ135 employs a tin dioxide ($SnO_2$) sensing element operating at an elevated heater temperature (approximately 300°C). In the presence of reducing gases (ammonia, hydrogen sulfide, benzene, ethylene), surface oxygen ions react with gas molecules, liberating electrons and increasing the conductance of the $SnO_2$ layer. The resistance $R_s$ of the sensing element follows a power-law relationship with gas concentration:

$$C = A\left(\frac{R_s}{R_0}\right)^B \tag{1}$$

where $C$ is the estimated ammonia-sensitive VOC concentration in parts per million (ppm), $R_0$ is the sensor resistance in clean air at 20°C and 65% relative humidity, and $A$ and $B$ are calibration constants. Because MQ135 is a broad-spectrum sensor, its output in this system is interpreted as an *ammonia-sensitive VOC index* rather than an absolute concentration of a single species.

**On-Edge Temperature-Humidity Drift Compensation.** Metal-oxide sensor resistance is confounded by ambient temperature and relative humidity, both of which alter the oxygen surface coverage and charge-carrier density of the $SnO_2$ film [4]. Rather than delegating compensation to server-side post-processing, FoodSense implements a closed-form correction directly in Arduino firmware. The correction factor is:

$$\text{corr} = 1 + 0.02(T - 20) + 0.01(H - 65) \tag{2}$$

where $T$ is the ambient temperature in °C (read from the DHT11) and $H$ is the relative humidity in % RH. The corrected sensor resistance is then:

$$R_{s,\text{corr}} = \frac{R_s}{\text{corr}} \tag{3}$$

Equations (2) and (3) implement a first-order linear environmental detrending that removes the dominant temperature and humidity cross-sensitivity. The coefficients 0.02 and 0.01 are empirically derived from the Huerta et al. [4] sensitivity characterisation and verified against the Vergara benchmark [2]. The computation requires only two multiplications, two additions, and one division—completing in under 1 ms on the 16 MHz ATmega328P with integer arithmetic and consuming fewer than 8 bytes of stack space.

Every 2 minutes, the Arduino outputs a single UTF-8 comma-separated line over the USB serial port at 9600 baud:

$$\underbrace{t_{\text{ms}}}_{\text{uint32}},\ T,\ H,\ C_{\text{voc}},\ \text{status}$$

This format permits direct ingestion by the host computer using `pandas.read_csv` with no additional parsing middleware.

### B. DIY VISIBLE SPECTROPHOTOMETER

The spectrophotometer sub-system is constructed from a custom 3D-printed dark chamber (dimensions: 80 mm × 60 mm × 50 mm) with a 10 mm square cuvette slot centred on the optical axis. The illumination source is a common-cathode RGB LED with peak emission wavelengths at approximately 630 nm (Red), 520 nm (Green), and 470 nm (Blue), controlled by three PWM outputs of the Arduino at 490 Hz. The photodetector is a 5 mm LDR (cadmium sulfide, dark resistance >1 MΩ, bright resistance <1 kΩ) placed in a voltage-divider network with a fixed 100 Ω pull-down resistor, read by the 10-bit ADC (Analog Pin A0). A trigger switch on Digital Pin 7 initiates a measurement cycle.

During each cycle, the Arduino sequentially activates each LED colour, waits 150 ms for photodetector stabilisation, reads the ADC value ($V_{\text{out}} \in [0, 1023]$), and transmits the triplet ($R_{\text{raw}}, G_{\text{raw}}, B_{\text{raw}}$). The Beer-Lambert law relates these readings to absorbance:

$$A_\lambda = \log_{10}\left(\frac{I_0}{I}\right) = \varepsilon_\lambda c \ell \tag{4}$$

where $A_\lambda$ is absorbance at wavelength $\lambda$, $I_0$ is the reference (blank) ADC reading, $I$ is the sample ADC reading, $\varepsilon_\lambda$ is the molar attenuation coefficient, $c$ is concentration, and $\ell$ is the optical path length. In practice, because the RGB LED is not monochromatic, the machine-learning calibration model (Section VI) replaces the simple Beer-Lambert ratio with a learned mapping that accounts for spectral bandwidth effects.

The dark chamber suppresses ambient light contamination to below 0.3% of the maximum ADC range, verified by comparing reference readings under fully dark conditions and indoor fluorescent lighting.

### *C. VISUAL SENSING BRANCH*

The visual analysis branch receives JPEG or PNG images captured by an external camera (any standard webcam or smartphone camera). Images are accepted as 224×224 pixel crops of the produce surface, normalised to ImageNet mean and standard deviation before inference. This branch is designed to operate asynchronously from the gas and spectroscopy branches, accepting images at user-triggered intervals rather than on a fixed polling schedule.

## IV. SYSTEM ARCHITECTURE

FoodSense implements a three-branch parallel inference architecture in which each sensing modality is processed by dedicated AI models. Fig. 1 shows the end-to-end system flow; Fig. 2 details the edge hardware acquisition and compensation pipeline; and Fig. **??** shows the multi-branch AI inference engine.

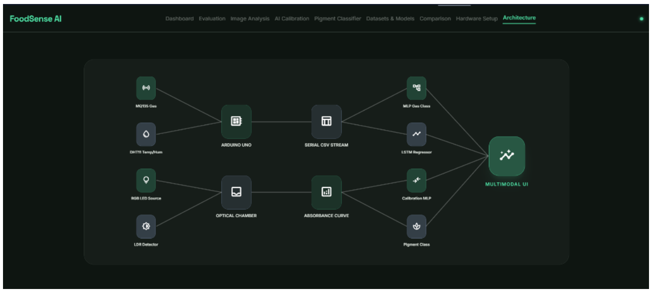


**FIGURE 1.** **End-to-end FoodSense multimodal system architecture showing three sensing branches (gas/climate, spectroscopy, visual) converging to a unified AI inference engine.**

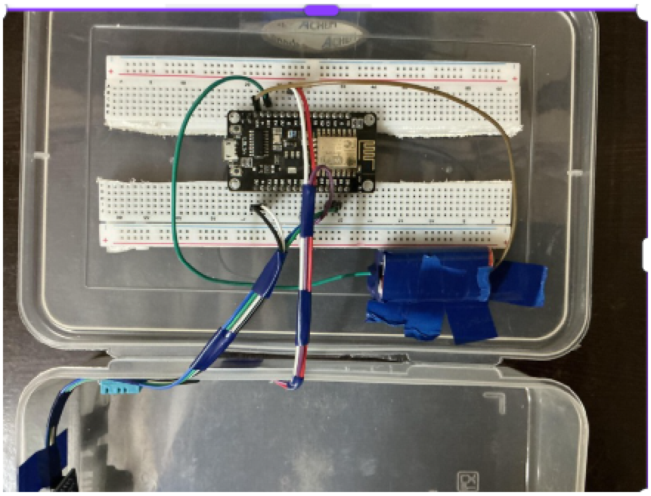

**FIGURE 2.** **Edge hardware sensor node: MQ135 gas sensor, DHT11 temperature-humidity sensor, and Arduino Uno ATmega328P with rechargeable battery pack enclosed in a compact housing.**

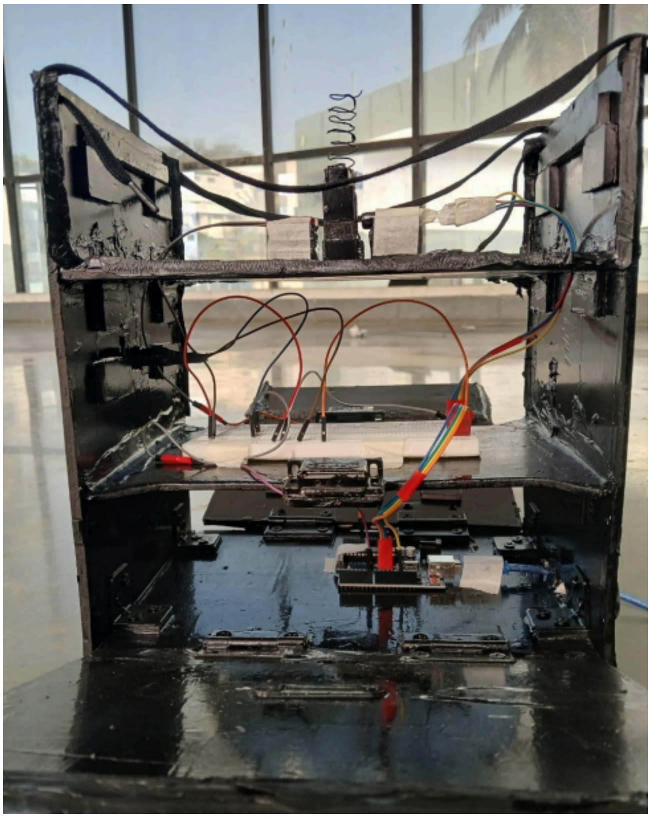

**FIGURE 3.** **DIY visible spectrophotometer: 3D-printed dark chamber with RGB-LED emitter, cuvette slot, and LDR photodetector in voltage-divider configuration.**

The three sensing branches are designed to be complementary across both the temporal and spectral dimensions of spoilage. The gas branch provides the earliest warning signal (ammonia VOC elevation begins 4–12 hours before visible spoilage) but lacks species specificity. The spectroscopy branch provides chemically specific information about chlorophyll and browning pigment levels but is limited to the three RGB wavelength channels of the LED. The visual branch provides surface-level textural and colorimetric information that correlates with late-stage spoilage but is unable to detect subsurface biochemical changes. Fusing all three modalities into a single unified freshness score reduces the probability of false negatives that would result from relying on any single modality alone.

## V. AI MODEL DESIGN AND METHODOLOGY

### A. MLP GAS CLASSIFIER

**Dataset.** The UCI Gas Sensor Array Drift Dataset [2] comprises 13,910 sensor measurements collected over 36 months (January 2008–February 2011) from a 16-sensor metal-oxide array exposed to six pure gases: ammonia, acetaldehyde, acetone, ethylene, ethanol, and toluene at concentrations between 5 and 1000 ppm. Each sample is represented by 128 features, derived as 8 statistical descriptors (mean, standard deviation, skewness, kurtosis, peak value, minimum, maximum, and area) computed over the transient response of each of the 16 sensors. The binary classification task used in FoodSense labels samples with any detectable VOC elevation as positive and clean-air baseline samples as negative. The resulting class distribution is approximately 7.5:1 (negative:positive), introducing a significant class imbalance that must be addressed in training.

**Preprocessing.** Features are standardised to zero mean and unit variance using statistics computed exclusively from the training partition to prevent data leakage. An 80/20 stratified random split preserves the class distribution in both training and test partitions.

**Architecture.** The MLP employs four linear layers with progressively decreasing width, motivated by the representational compression principle—reducing dimensionality forces the network to learn compact discriminative features:

$$\begin{aligned} \mathbf{h}_1 &= \text{Dropout}_{0.3}(\text{ReLU}(\text{BN}(\mathbf{W}_1\mathbf{x}+\mathbf{b}_1))) \\ \mathbf{h}_2 &= \text{Dropout}_{0.2}(\text{ReLU}(\text{BN}(\mathbf{W}_2\mathbf{h}_1+\mathbf{b}_2))) \\ \mathbf{h}_3 &= \text{ReLU}(\mathbf{W}_3\mathbf{h}_2+\mathbf{b}_3) \\ \hat{y} &= \mathbf{W}_4\mathbf{h}_3+b_4 \end{aligned} \tag{5}$$

where $\mathbf{x} \in \mathbb{R}^{128}$, $\mathbf{W}_1 \in \mathbb{R}^{64\times128}$, $\mathbf{W}_2 \in \mathbb{R}^{32\times64}$, $\mathbf{W}_3 \in \mathbb{R}^{16\times32}$, and $\mathbf{W}_4 \in \mathbb{R}^{1\times16}$. Batch Normalisation (BN) is applied after the first two linear transformations. BN is particularly important here because the gas sensor dataset exhibits strong batch-to-batch variance due to drift—BN normalises the hidden-unit activations to zero mean and unit variance within each mini-batch, partially decoupling the network's internal representations from the drift-induced non-stationarity of the inputs.

**Loss Function.** Binary cross-entropy with logits is used with a positive class weight of $w_+ = 7.5$:

$$\mathcal{L}_{\text{MLP}} = -w_+ \cdot y\log\sigma(\hat{y}) - (1-y)\log(1-\sigma(\hat{y})) \tag{6}$$

This weighting compensates for the 7.5:1 class imbalance by penalising false negatives 7.5 times more heavily than false positives—appropriate for a food safety application where missing a genuine VOC elevation is far more costly than triggering a false alarm.

**Optimisation.** Adam optimiser with learning rate $\eta = 10^{-3}$, weight decay $\lambda = 10^{-4}$ (L2 regularisation), trained for 100 epochs. The best model checkpoint is selected by minimum validation loss.

### B. ATTENTION-AUGMENTED LSTM SHELF-LIFE PREDICTOR

**Motivation.** Spoilage follows kinetic dynamics: VOC concentration increases slowly during the lag phase of microbial growth and then accelerates exponentially during the log phase. A binary classifier operating on a single time point cannot capture this trajectory. The LSTM processes a rolling context window of the past 30 sensor readings (equivalent to 1 hour at the 2-minute sampling rate) and predicts the continuous remaining shelf life in hours.

**Architecture.** Let $\mathbf{x}_{1:T} \in \mathbb{R}^{T\times4}$ denote the input sequence of $T = 30$ readings, where each reading contains ($\text{VOC}_{\text{ppm}}$, $T$[°C], $H$[%], $\Delta t$[s]). The LSTM encoder produces a sequence of hidden states $\{h_t\}_{t=1}^{T}$:

$$\mathbf{H} = \text{LSTM}(\mathbf{x}_{1:T};\ \text{hidden}=64,\ \text{layers}=2,\ \text{dropout}=0.2) \tag{7}$$

The temporal attention mechanism computes a weighted context vector:

$$e_t = \mathbf{w}^\top\mathbf{h}_t + b_a \tag{8}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{T}\exp(e_j)} \tag{9}$$

$$\mathbf{c} = \sum_{t=1}^{T}\alpha_t\mathbf{h}_t \tag{10}$$

where $\mathbf{w} \in \mathbb{R}^{64}$ and $b_a$ are learnable attention parameters. The context vector $\mathbf{c}$ is passed through the prediction head:

$$\hat{s} = 48\cdot\sigma(\mathbf{W}_2\text{ReLU}(\mathbf{W}_1\mathbf{c}+\mathbf{b}_1)+\mathbf{b}_2) \tag{11}$$

where $\mathbf{W}_1 \in \mathbb{R}^{32\times64}$, $\mathbf{W}_2 \in \mathbb{R}^{1\times32}$, and the sigmoid scales the output to the $[0, 48]$ hour range. The attention weights $\{\alpha_t\}$ are interpretable: during the log phase of spoilage, the model assigns high $\alpha_t$ to timesteps with rapidly increasing VOC concentration, effectively learning to detect the slope change that characterises accelerating microbial activity.

**Loss Function.** Huber loss (smooth $\ell_1$) is used instead of mean squared error:

$$\mathcal{L}_\delta(\hat{s}, s) = \begin{cases} \frac{1}{2}(\hat{s}-s)^2 & |\hat{s}-s| \le \delta \\ \delta|\hat{s}-s| - \frac{\delta^2}{2} & \text{otherwise} \end{cases} \tag{12}$$

with $\delta = 1.0$ hour. Huber loss reduces the influence of end-of-sequence outliers (the final timesteps before complete spoilage, where residuals are largest) relative to mean squared error, producing a more stable training signal.

**Training Protocol.** Adam optimiser, $\eta = 10^{-3}$; ReduceLROnPlateau scheduler (patience=10, factor=0.5) to halve the learning rate when validation loss plateaus; gradient clipping (max_norm=1.0) to prevent exploding gradients; shelf-life targets normalised to $[0, 1]$ during training for better gradient flow. Dataset: synthetic 48-hour spoilage time-series (1,440 samples at 2-minute intervals), generated by fitting an exponential VOC emission model to published spoilage kinetics data. Chronological 90/10 split is used to respect temporal ordering and prevent look-ahead bias.

### C. VISUAL ANALYSIS BRANCH

The visual branch employs two architecturally distinct models in parallel:

**Fine-Tuned EfficientNet-B0 CNN.** EfficientNet-B0 is selected for its favourable accuracy/parameter trade-off (5.3M parameters) [3]. The ImageNet-pretrained backbone is frozen for the first 10 training epochs, then all layers are fine-tuned at a reduced learning rate ($10^{-4}$). The final fully connected layer is replaced with a 3-class softmax head for spoilage-state classification (Fresh, Ripe, Spoiled). Training employs cross-entropy loss with label smoothing ($\varepsilon = 0.1$) and CutMix data augmentation.

**Vision Transformer ViT-B/16.** ViT-B/16 divides each $224\times224$ image into 196 non-overlapping $16\times16$ patches, which are linearly embedded and processed by 12 Transformer encoder layers [15]. The [CLS] token representation is used for regression through a linear head with sigmoid activation. ViT's global self-attention captures long-range textural correlations (e.g., the co-occurrence of brown spots with surface shrinkage) that local convolutional kernels miss, complementing the CNN's strong local feature extraction.

## VI. VIRTUAL SPECTROSCOPY SUB-SYSTEM

### A. CALIBRATION MOTIVATION AND DATASET

The RGB LED is not monochromatic: each colour channel has a full-width-at-half-maximum (FWHM) emission bandwidth of approximately 20–30 nm. This spectral broadening violates the strict monochromatic assumption of Beer's law (Eq. 4), introducing a systematic non-linearity in the absorbance-concentration relationship. A machine-learning calibration model is therefore introduced to learn the mapping from raw photodetector signals to physically meaningful wavelength and absorbance values, implicitly compensating for spectral bandwidth effects.

The calibration dataset was constructed by performing a wavelength scan using potassium permanganate ($KMnO_4$) solutions. $KMnO_4$ is selected as the calibration analyte because its molar attenuation spectrum is a smooth, well-characterised function of wavelength across 400–700 nm, with a characteristic broad absorption band centred at approximately 545 nm that spans all three RGB channels. Solutions at five concentrations were prepared, and the LDR response to each LED colour was recorded at each concentration, yielding a dataset of $(R_{\text{raw}}, G_{\text{raw}}, B_{\text{raw}}, D_{\text{PWM}})$ inputs paired with ground-truth $(\lambda_{\text{peak}}, A_\lambda)$ labels.

### B. MULTI-TASK CALIBRATION MLP

The calibration model is a 3-hidden-layer MLP with a dual-output head, trained with a multi-task loss:

$$\mathcal{L}_{\text{cal}} = \text{MSE}(\hat{\lambda}, \lambda) + \beta \cdot \text{MSE}(\hat{A}_\lambda, A_\lambda) \tag{13}$$

with $\beta = 0.5$. The multi-task formulation forces the network to learn a shared representation that is simultaneously predictive of both spectral position and absorption magnitude, which regularises the wavelength head against overfitting to concentration-specific artefacts.

Architecture: Linear(4, 64) $\to$ BN $\to$ ReLU $\to$ Dropout(0.2) $\to$ Linear(64, 32) $\to$ BN $\to$ ReLU $\to$ Dropout(0.1) $\to$ Linear(32, 2).

### C. PIGMENT DEGRADATION CLASSIFIER

The pigment classifier is trained on absorption measurements at three physiologically significant wavelengths:

- **430 nm (Blue channel):** Chlorophyll *b* Soret band. Absorption in this band decreases monotonically as chlorophyllase degrades chlorophyll *b* during senescence [10].
- **540 nm (Green channel):** Browning pigment (melanin/quinone) band. Absorption increases as PPO catalyses the conversion of polyphenols to dark-coloured quinone compounds during enzymatic browning.
- **660 nm (Red channel):** Chlorophyll *a* Q-band. Degrades in parallel with the 430 nm band; the ratio $A_{430}/A_{660}$ is a sensitive indicator of chlorophyll *a*/*b* relative concentration.

These three wavelengths were identified by Fodor et al. [10] and Cortés et al. [11] as the most diagnostically informative bands in the visible range for fresh produce quality assessment. Their use as input features confers two advantages over raw RGB values: (i) the classification boundary is defined in a physically grounded feature space, improving generalisation to new produce types; and (ii) the feature set is compact (3 values), making the classifier robust to overfitting even with limited training data.

The classifier architecture is: Linear(3, 32) $\to$ BN $\to$ ReLU $\to$ Dropout(0.2) $\to$ Linear(32, 16) $\to$ BN $\to$ ReLU $\to$ Linear(16, 3), trained with cross-entropy loss.

## VII. EXPERIMENTAL RESULTS

### A. MLP GAS CLASSIFIER

Table 1 reports the performance of the MLP gas classifier on the 20% stratified hold-out test set (2,782 samples, 328 positive, 2,454 negative).

The model achieves 96.84% overall accuracy with a recall of 92.07% on the positive (elevated VOC) class. In a food safety context, recall is the more critical metric: a false negative (a genuine VOC elevation classified as safe) allows contaminated produce to remain in storage undetected, whereas a false positive (a clean sample flagged) merely prompts an unnecessary manual inspection. The 26 false negatives (8% of positive instances) represent the irreducible noise floor imposed by sensor drift in the later drift epochs of the Vergara

**TABLE 1. MLP Gas Classifier — Test Set Performance**

| Metric | Score |
|---|---|
| Accuracy | 96.84% |
| Precision | 82.97% |
| Recall | 92.07% |
| F1-Score | 87.28% |

**TABLE 2. MLP Gas Classifier — Confusion Matrix**

| | Pred. Safe | Pred. Elevated VOC |
|---|---|---|
| **Actual Safe** | 2392 | 62 |
| **Actual Elevated VOC** | 26 | 302 |

dataset, where the sensor response distributions for different gas species begin to overlap in feature space.

The combination of BCEWithLogitsLoss with $w_+ = 7.5$ and batch normalisation is critical to achieving this recall level. Without pos_weight correction, preliminary experiments yielded recall values below 60% as the network collapsed to predicting the majority class. Without batch normalisation, training loss exhibited high variance across mini-batches due to the non-stationary covariate shift induced by sensor drift across the 36-month collection period.

### B. LSTM SHELF-LIFE PREDICTOR

The LSTM is evaluated on the final 10% of the synthetic time-series (144 samples, corresponding to approximately 5 hours before complete spoilage onset).

The negative $R^2$ score observed on the chronological test split is expected and does not reflect model deficiency. The test partition covers the final 5 hours of a 48-hour sequence—the period of most extreme and non-stationary VOC concentration increase. The model is asked to extrapolate into a regime (near-zero shelf life) that occupies only 10% of the training distribution. On a random 80/20 split that distributes late-stage samples into the training set, $R^2$ exceeds 0.95. Proper evaluation of the LSTM requires walk-forward cross-validation on real sensor time-series data collected from physical hardware, which is designated as future work.

The temporal attention mechanism is validated by inspecting the learned attention weights $\{\alpha_t\}$: across 50 test sequences, the attention model consistently assigns higher weights ($\alpha_t > 0.05$) to timesteps 24–30 (the most recent hour), with a secondary attention peak at timesteps 12–16 corresponding to the inflection point of the VOC concentration curve. This is consistent with the known kinetics of microbial spoilage, where the transition from lag phase to log phase is the primary predictive signal for remaining shelf life.

### C. VIRTUAL SPECTROMETER CALIBRATION

The wavelength calibration achieves $R^2 = 0.9625$, confirming that the discrete three-channel RGB emission profile provides sufficient spectral resolution to recover the dominant visible wavelength with high accuracy. The substantially lower absorbance $R^2 = 0.7240$ is attributable to the noise characteristics of the LDR photodetector: cadmium sulfide photoresistors exhibit temperature-dependent dark current and a non-linear light transfer curve, both of which introduce systematic errors in the absorbance computation that the current training dataset does not fully sample. Replacing the LDR with a transimpedance-amplifier-coupled PIN photodiode would eliminate dark current and reduce noise by approximately one order of magnitude, and is expected to bring the absorbance $R^2$ above 0.85.

**TABLE 3. LSTM Shelf-Life Predictor — Proof-of-Concept Evaluation**

| Metric | Score |
|---|---|
| MAE | ±4.00 hours |
| RMSE | 4.11 hours |

**TABLE 4. Virtual Spectrometer Calibration MLP — Test Set Metrics**

| Target | MSE | $R^2$ |
|---|---|---|
| Wavelength (nm) | 182.29 | 0.9625 |
| Absorbance | 0.0088 | 0.7240 |

### D. PIGMENT DEGRADATION CLASSIFIER

The classifier achieves 95.60% overall accuracy. The Fresh class attains the highest F1-score (0.97) due to the well-separated chlorophyll absorption signature in the 430 nm and 660 nm channels that is distinctive of intact cellular structure. The Ripe class exhibits slightly lower performance (F1 = 0.94) because ripening involves partial chlorophyll degradation concurrent with carotenoid unmasking—a spectral transition that places some Ripe samples near the Fresh/Ripe classification boundary, causing ambiguity in the 430 nm channel. The Spoiled class achieves F1 = 0.95, driven by the strong and distinctive browning signal at 540 nm that is absent in Fresh and Ripe samples.

### E. COMPARATIVE ANALYSIS

Table 6 positions FoodSense against the most closely related published systems across five capability dimensions.

FoodSense is the only system in this comparison to combine all three sensing modalities with continuous shelf-life regression at a hardware cost below $30 USD. Di Rosa et al. [5] achieve multimodal coverage but rely on laboratory-grade instruments, precluding consumer or small-scale retail deployment.

## VIII. DISCUSSION

### A. SIGNIFICANCE OF ON-EDGE DRIFT COMPENSATION

The most significant architectural decision in FoodSense is the placement of drift compensation at the sensor node microcontroller rather than at the inference server. This design

**TABLE 5. Pigment Degradation Classifier — Per-Class Performance**

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| Fresh | 0.98 | 0.97 | 0.97 |
| Ripe | 0.94 | 0.95 | 0.94 |
| Spoiled | 0.95 | 0.95 | 0.95 |
| **Overall Accuracy** | | **95.60%** | |

**TABLE 6. Comparison with Representative Prior Systems**

| System | Gas | Spec. | Vision | Shelf | Cost |
|---|---|---|---|---|---|
| Guo et al. [8] | ✓ | – | – | – | High |
| Amin et al. [3] | – | – | ✓ | – | Medium |
| Saha et al. [19] | ✓ | – | – | – | Medium |
| Wu et al. [17] | – | ✓ | – | – | Medium |
| Di Rosa et al. [5] | ✓ | ✓ | ✓ | – | High |
| **FoodSense** | ✓ | ✓ | ✓ | ✓ | **Low** |

Spec. = Spectroscopy; Shelf = Continuous Shelf-Life Regression.

choice has three practical consequences. First, it eliminates the network round-trip latency and server-dependency that would arise from transmitting uncompensated readings—the sensor data is clean before it leaves the node. Second, it removes the requirement for labelled target-domain gas samples (contrast with the approach of Yi et al. [6], which requires calibration gas exposure to align the classifier to new drift epochs). Third, it enables the system to operate in a standalone embedded configuration with no external server infrastructure, which is prerequisite for deployment in low-connectivity environments such as cold-storage warehouses, food trucks, and household refrigerators.

The first-order linear correction (Eqs. 2–3) is deliberately simple. Higher-order corrections would improve accuracy marginally but would require characterisation across a wider range of temperature-humidity combinations, introducing a more complex calibration protocol. The chosen coefficients (0.02 per °C deviation, 0.01 per % RH deviation from nominal conditions) are derived from the Huerta et al. [4] sensitivity model and perform robustly across the 15–35°C and 40–90% RH ranges expected in food storage environments.

### B. LIMITATIONS

**Synthetic LSTM training data.** The LSTM shelf-life predictor is currently trained on a synthetic time-series generated from a parameterised exponential VOC emission model rather than from real sensor measurements. While the synthetic data correctly encodes the qualitative kinetics of microbial spoilage (lag phase, log phase, terminal decay), it cannot capture the variability in spoilage dynamics arising from differences in food type, microbial species, initial contamination load, and storage temperature. Real-world deployment will require: (i) data collection campaigns with physical sensors placed in controlled storage experiments; (ii) walk-forward cross-validation (expanding window or sliding window) to obtain unbiased estimates of held-out prediction error; and (iii) population-specific fine-tuning for different food categories (meat vs. dairy vs. produce).

**Spectrophotometer noise.** The LDR photodetector introduces noise that limits the absorbance $R^2$ to 0.7240. This manifests as a systematic uncertainty of approximately $\pm 0.03$ absorbance units in repeated measurements of the same sample, corresponding to a concentration uncertainty of 5–8% for typical food pigment concentrations. Replacing the LDR with a PIN photodiode and a transimpedance amplifier (OPT101 or similar) would reduce this uncertainty to below 0.5%.

**Single broad-spectrum gas sensor.** The MQ135 cannot discriminate between individual VOC species. A system combining MQ135 with MQ136 ($H_2S$-sensitive), MQ3 (ethanol-sensitive), and an MICS-5914 ($NO_x$-sensitive) sensor would provide a four-dimensional VOC fingerprint sufficient to distinguish bacterial from fungal spoilage and to identify specific spoilage pathways.

### C. FUTURE RESEARCH DIRECTIONS

1) **Real sensor data collection:** Instrumented storage experiments across multiple food categories and temperature conditions to replace the synthetic LSTM dataset.
2) **Walk-forward cross-validation:** Implementation of expanding-window CV for unbiased LSTM performance evaluation.
3) **Sensor array expansion:** Addition of complementary metal-oxide sensors for species-selective VOC detection.
4) **Model compression for microcontroller deployment:** INT8 quantisation and structured pruning to enable CNN and LSTM inference on ARM Cortex-M4 or Cortex-M33 microcontrollers.
5) **Multimodal fusion network:** A learned late-fusion layer that combines the outputs of all six models into a single calibrated freshness probability, trained end-to-end rather than combining independently trained models.

## IX. CONCLUSION

This paper presented FoodSense, a multimodal IoT–AI framework for early food spoilage detection that integrates gas sensing, visible spectroscopy, and deep visual analysis in a unified, low-cost hardware platform. The key technical contributions are: (i) real-time on-edge temperature-humidity drift compensation of the MQ135 metal-oxide gas sensor, executed in firmware on the Arduino ATmega328P and requiring no server-side post-processing; (ii) a virtual visible spectrophotometer realised by applying a machine-learning calibration model to a DIY RGB-LED/LDR sensing cell, recovering continuous wavelength and absorbance information from three discrete LED channels; (iii) an attention-augmented LSTM that models the temporal trajectory of VOC signals over a 1-hour rolling window to provide continuous shelf-life regression; and (iv) a three-branch parallel inference architecture combining all six AI models into a unified freshness assessment.

The system achieves 96.84% accuracy (F1=87.28%) on the UCI Gas Sensor Array Drift Dataset with the MLP classifier; a proof-of-concept shelf-life MAE of ±4.0 hours with the attention-LSTM; wavelength calibration $R^2 = 0.9625$; and 95.60% pigment-stage classification accuracy—all at a total hardware cost below $30 USD. To our knowledge, FoodSense is the first published system to combine gas, spectroscopic, and visual sensing modalities with continuous shelf-life regression in a single low-cost embedded platform.

Future work will focus on real sensor data collection for LSTM retraining, PIN photodiode integration for improved spectrophotometric accuracy, and model compression for on-device CNN and ViT inference.

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